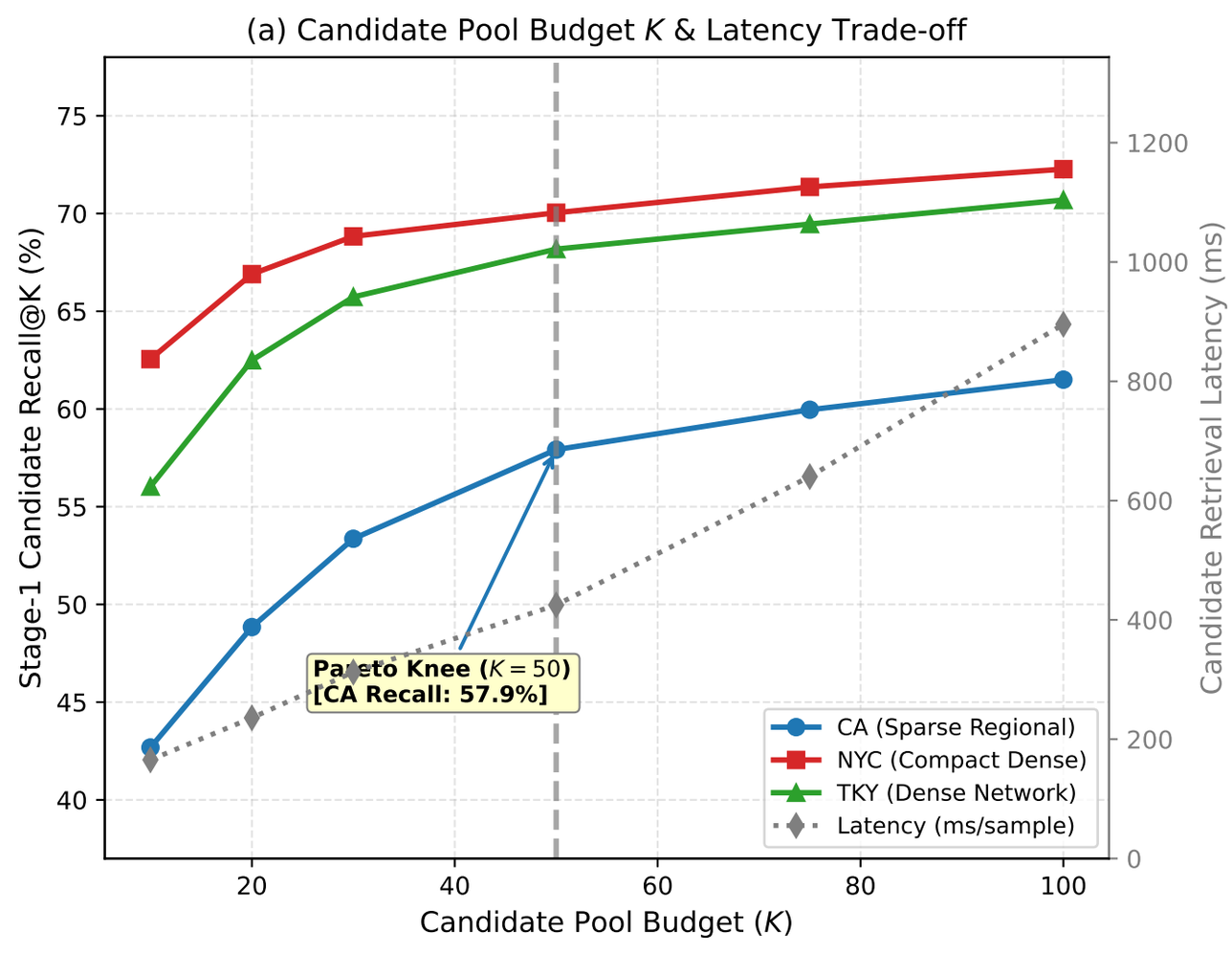
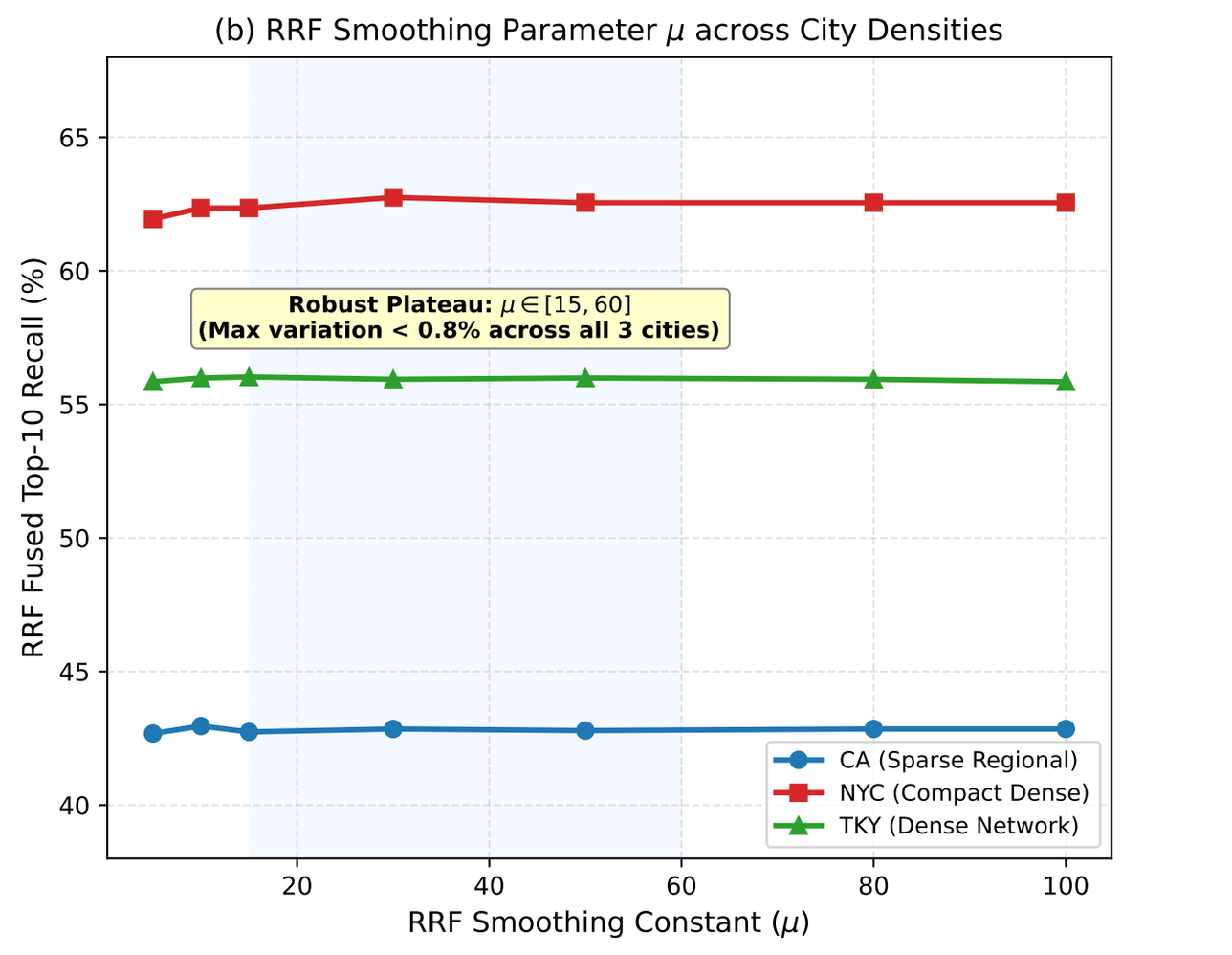
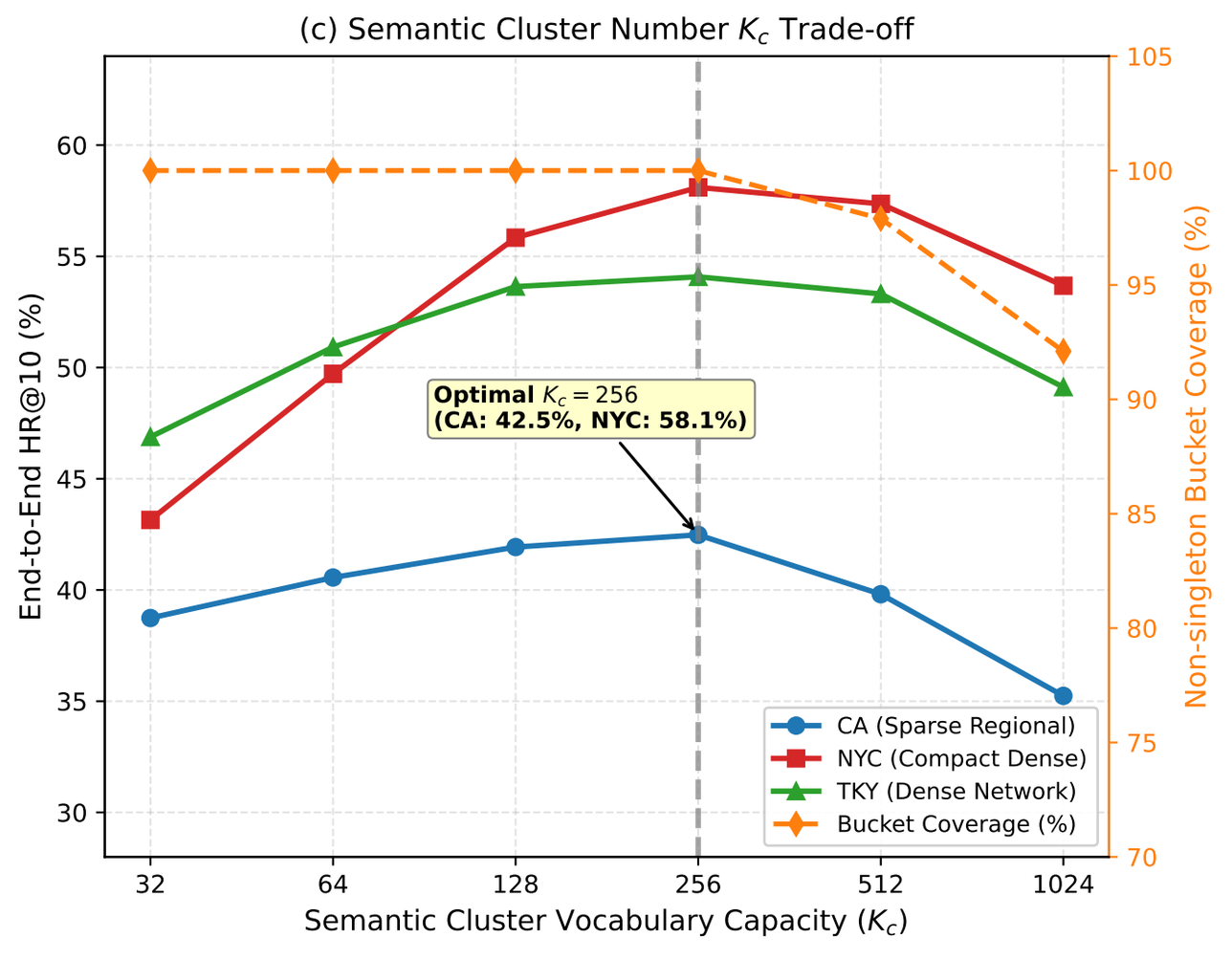
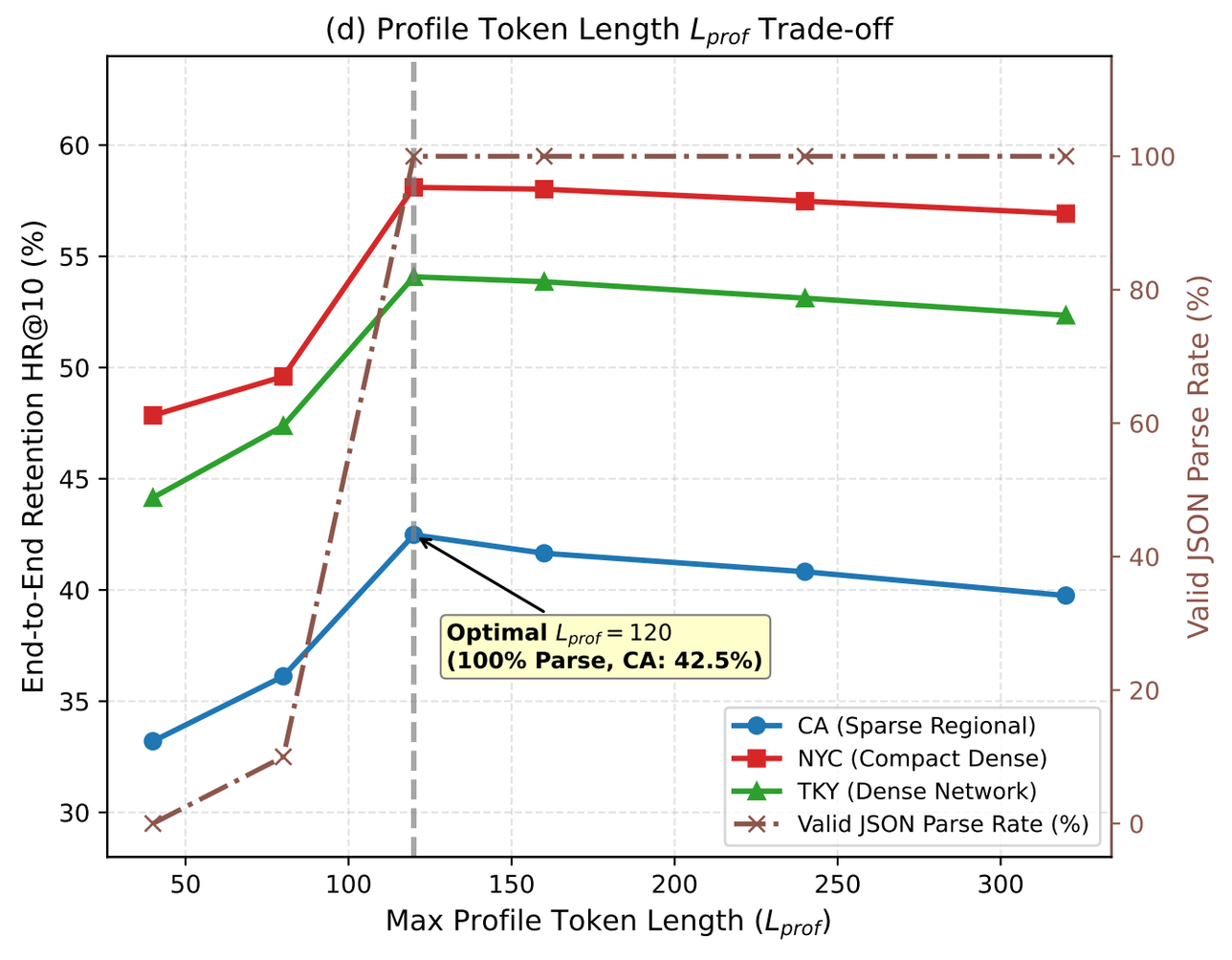

(a) Candidate Pool Budget K & Latency Trade-off
Stage-1 Candidate Recall@K (%)
Candidate Retrieval Latency (ms)
Candidate Pool Budget (K)
Pareto Knee ($K = 50$)
[CA Recall: 57.9%]
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)


(b) RRF Smoothing Parameter μ across City Densities
RRF Fused Top-10 Recall (%)
RRF Smoothing Constant (μ)
Robust Plateau: $\mu \in [15, 60]$
(Max variation < 0.8% across all 3 cities)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)


(c) Semantic Cluster Number K_c Trade-off
End-to-End HR@10 (%)
Non-singleton Bucket Coverage (%)
Semantic Cluster Vocabulary Capacity (K_c)
Optimal $K_c = 256$
(CA: 42.5%, NYC: 58.1%)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Bucket Coverage (%)


(d) Profile Token Length L_{prof} Trade-off
End-to-End Retention HR@10 (%)
Valid JSON Parse Rate (%)
Max Profile Token Length (L_{prof})
Optimal $L_{prof} = 120$
(100% Parse, CA: 42.5%)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Valid JSON Parse Rate (%)